

# Compiled with problems:

## ERROR in ./docs/semester-1/gqur-300/unit-03/topic-01.mdx

Module build failed (from ./node\_modules/@docusaurus/mdx-loader/lib/index.js):

Error: MDX compilation failed for file

["/home/a2ahs/mega\\_book\\_for\\_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx"](/home/a2ahs/mega_book_for_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx)

Cause: Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a letter, `\$`, or `\_`

Details:

```
{
  "column": 36,
  "file": "",
  "message": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "line": 141,
  "name": "141:36",
  "place": {
    "_bufferIndex": 33,
    "_index": 0,
    "line": 141,
    "column": 36,
    "offset": 7642
  },
  "reason": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "ruleId": "unexpected-character",
  "source": "micromark-extension-mdx-jsx",
  "url": "https://github.com/micromark/micromark-extension-
mdx-jsx#unexpected-character-at-expected-expect"
}
```

at compileToJSX

decimal column of the same sale. His new assistant is confused: "Are these all numbers? One has a minus, one has a bar, one has a dot." Mr. Rashid laughs and draws a picture on the back of an invoice: the number family tree. All four entries are numbers; they simply live in different rooms of the same house. By the end of this topic you will be able to draw that picture yourself, and more importantly, compute across it.

# Explanation

## Whole numbers and integers

Whole numbers are the counting numbers and zero: 0, 1, 2, 3, and so on. They answer "how many?" when the answer cannot be negative: 240 pupils, 16 groups, 5 books. Integers extend whole numbers below zero: ..., -3, -2, -1, 0, 1, 2, 3, ... The negative side is not strange; it is bookkeeping made honest. A debt of 3,500 rupees is written -3,500 so that adding it to the cash box gives the true position. Temperature in Skardu in December, floors below ground level, marks deducted for wrong answers: all are natural uses of negative numbers.

Two everyday situations build integer sense:

- **Debt and credit:** cash box 8,000 rupees, debt 3,500 rupees; the position is  $8,000 + (-3,500) = 4,500$  rupees.
- **Change and difference:** the temperature rose from -2 to 6 degrees; the change is  $6 - (-2) = 8$  degrees, not 4.

The National Curriculum Council publishes the national mathematics curriculum for Grades 1-12 (NCC, n.d.); whole numbers and integers sit at its foundation, as they do in every school curriculum, because everyday bookkeeping demands them.

## Fractions and decimals

| Operation             | With fractions                                                              | With decimals           |
|-----------------------|-----------------------------------------------------------------------------|-------------------------|
| <b>Addition</b>       | $\frac{1}{2} \text{ m} + \frac{1}{4} \text{ m} = \frac{3}{4} \text{ m}$     | $0.5 + 0.25 = 0.75$     |
| <b>Subtraction</b>    | $1 \text{ m} - \frac{1}{4} \text{ m} = \frac{3}{4} \text{ m}$               | $1.00 - 0.25 = 0.75$    |
| <b>Multiplication</b> | $\frac{1}{2} \text{ of } \frac{1}{2} \text{ m} = \frac{1}{4} \text{ m}$     | $0.5 \times 0.5 = 0.25$ |
| <b>Division</b>       | $\frac{1}{2} \text{ m split by } \frac{1}{4} \text{ m} = 2 \text{ lengths}$ | $0.50 / 0.25 = 2$       |

Same metre of cloth, two notations: every fraction has a decimal name.

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A fraction names a part of a whole or a share of a group: three quarters means 3 of 4 equal parts. The bottom number (the denominator) counts the equal parts; the top number (the numerator) counts the parts taken. A decimal is the same idea written in place value: 0.75 says 7 tenths and 5 hundredths. Every fraction has a decimal name and almost every decimal in daily life is a fraction in disguise; the shop's "three quarter metres" and "0.75" were the same quantity twice.

The table above shows the four operations with fractions and decimals side by side, each with a metre-and-rupee example. Study one row at a time, then cover the answers and reproduce it.

**Worked example.** A tailor has 5.5 metres of cloth. A shalwar kameez suit needs 2.75 metres. How many suits, and what remains? Plan: divide. Carry out: 5.5 divided by 2.75 is exactly 2. Look back: 2 suits use 5.5 metres, so nothing remains - and the answer is reasonable because 5.5 is a bit more than two lots of 2.5. Notice how Unit 1's four-step strategy carries straight into computation.

**A common misconception to correct.** Many trainees believe "multiplying always makes bigger". With whole numbers it usually does, but  $12 \times \frac{1}{2}$  is 6 (smaller), and  $12 \times 0.1$  is 1.2 (much smaller). Multiplying by a number between

0 and 1 shrinks. This one idea explains dozens of daily errors, from fabric orders to electricity unit calculations. Test every answer against your number sense: could this result be sensible?

## Activity: The bazaar ledger

Work in pairs. Your tutor gives each pair a mock shop ledger with six entries mixing whole numbers, signed amounts, fractions and decimals (for example: sold 3.25 kg sugar; loan repaid -1,200 rupees;  $\frac{1}{2}$  kg tea received). First, place every entry on a number line drawn on paper. Then compute the shop's closing position. Finally, convert each fraction entry to its decimal form and check that your closing position does not change. Compare ledgers with the pair beside you. About 25 minutes.

## Check your understanding

1. Place these on a number line and order them from smallest to largest: 2, -3,  $\frac{1}{2}$ , 0.75, -0.5. (*Apply*)
2. A shop's cash box holds 9,000 rupees and its debts total 4,500 rupees. Write the shop's position using integers and compute it. (*Apply*)
3. A recipe needs  $\frac{3}{4}$  kg of flour and you have 1.5 kg. How many full recipes can you make, and what is left over? Show the division. (*Apply*)
4. In your own words, explain why  $12 \times 0.1$  is smaller than 12, using the idea of "part of a whole". (*Understand*)

## Summary

Whole numbers count what cannot be negative; integers add the negative side and make bookkeeping honest; fractions and decimals are two notations for parts of wholes, and every fraction has a decimal name. Operations work the same across the system, and the reasonableness check from Unit 1 applies to every result: multiplying by a number between 0 and 1 shrinks, and a negative result must mean something real in the situation.

# Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can place whole numbers, integers, fractions and decimals on one number line and order them.
- ☐ I can compute with integers in debt-and-credit situations.
- ☐ I can add, subtract, multiply and divide with fractions and decimals in daily-life examples, and convert between the two notations.

## Try this at your practicum school

Ask the school clerk (or your cooperating teacher) how the stationery register records part-used items (half reams of paper, part boxes of chalk). Note two entries that use fractions or decimals, copy them exactly, and convert each to the other notation. Bring the note to class.

## Summative task

A school canteen buys flour in 5.5 kg bags at 1,200 rupees per bag. One batch of parathas uses 0.75 kg. In a short written answer: (a) compute how many full batches one bag gives and the flour left over, (b) compute the flour cost per batch to the nearest rupee, and (c) check both answers for reasonableness using estimation, showing your check.

**Mini-rubric** (each criterion 0-2 marks):

- *Division correct*: full batches and remainder from 5.5 divided by 0.75 (7 batches, 0.25 kg left).
- *Cost per batch*: cost per batch computed with clear working (1,200 divided by 7.33 batches gives about 164 rupees; 1,200 divided by 7 full batches gives about 171 rupees treating the leftover as waste; either defensible computation earns the mark).
- *Reasonableness check*: an explicit estimation check of at least one answer, not a bare restatement. **(Analyze)**

# Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 1 Whole Numbers; Ch. 3 Integers; Ch. 4 Fractions; Ch. 5 Decimals). OpenStax. <https://openstax.org/details/books/prealgebra-2e>
- National Curriculum Council, Ministry of Federal Education and Professional Training, Government of Pakistan. *Mathematics: Suggested Guidelines (Grades 1-8)* and *NCP Mathematics Progression Grid (Grades 1-12)*. <https://ncc.gov.pk/Detail/ZjgzYzg2MmMtZDc0Zi00NjEzLTk5ZmYtZGJiNjc5ODljOGUx>

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